

Mobile App Developer

Step 2: Choose Android or iOS

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Choosing a platform is a product and career decision, not merely an IDE preference. Native Android development is centered on the Android platform and commonly Kotlin/Android Studio; native iOS development is centered on Apple's SDKs and Swift/Xcode. The platforms differ in UI frameworks, lifecycle APIs, distribution, signing and device ecosystems.

For a beginner, choosing one primary platform reduces context switching. Learn the native platform concepts even if you later use Flutter or React Native: application lifecycle, navigation, permissions, storage, background work, networking and platform security.

Your decision should consider the audience you want to build for, the devices you can test on, your operating system constraints, and the kinds of roles you want.

Core Concepts — Detailed Breakdown

- Android: application components, lifecycle, activities, navigation, Jetpack/Compose concepts, permissions, storage, networking and Play distribution.
- iOS: Swift, SwiftUI/UIKit concepts, view lifecycle, navigation, permissions, persistence, networking and App Store distribution.
- Development environment: Android tooling is cross-platform; standard iOS development requires Apple's macOS/Xcode workflow.
- Device fragmentation: Android has wider hardware variation; iOS has a narrower device matrix but strict platform conventions.
- Publishing: both platforms require signing, metadata, privacy considerations, release testing and store review.
- Cross-platform choice: Flutter and React Native can share application code but still expose platform-specific behavior and APIs.

What You Should Be Able To Do

- Install the official SDK/IDE and run a starter application.
- Create a three-screen prototype with navigation.
- List five platform-specific APIs you expect to learn later.
- Explain why you selected the platform.

Hands-on Project / Practice

- Create a small native app with a list screen, detail screen and form. Add one permission or platform API so you learn the platform boundary.

Recommended References

- Android Developers — <https://developer.android.com/docs>
- Apple Developer Documentation — <https://developer.apple.com/documentation/>
- Flutter Learn — <https://docs.flutter.dev/learn>
- React Native Documentation — <https://reactnative.dev/docs/getting-started>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.